

Food Classification using Deep Learning

Project Overview

- Deep Learning-based image classification system for food recognition.
- Uses Custom CNN and Transfer Learning models (VGG16, ResNet50).
- Designed for real-time prediction via web interface.

Project Highlights

- Custom CNN Architecture Implementation
- Transfer Learning using VGG16 and ResNet50
- Data Augmentation for improved generalization
- High accuracy multi-class classification
- Deployment ready with Streamlit / Flask
- Cloud compatible (Render)

Models Implemented

- Custom CNN: Convolution, MaxPooling, Flatten and Dense layers.
- VGG16: Pretrained on ImageNet for strong feature extraction.
- ResNet50: Deep residual network solving vanishing gradient problem.

Tech Stack

- Python
- TensorFlow / Keras
- NumPy
- OpenCV
- Matplotlib
- Streamlit / Flask
- Git & GitHub

Model Training Details

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Batch Size: 32
- Epochs: 20-30
- Evaluation Metric: Accuracy

Future Improvements

- Increase dataset size and categories
- Add calorie estimation feature
- Integrate recommendation system
- Enhance UI with better visualization

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